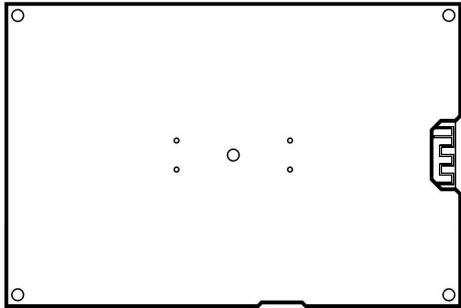
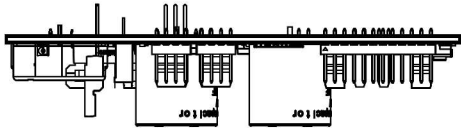
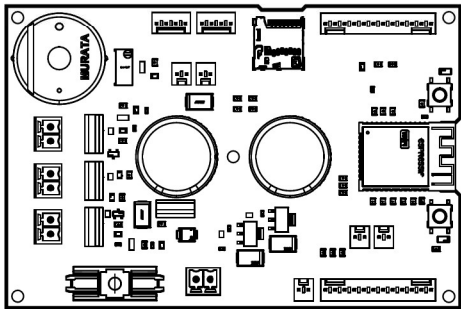
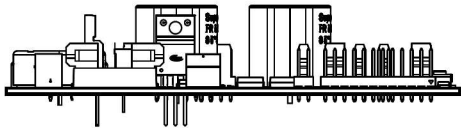
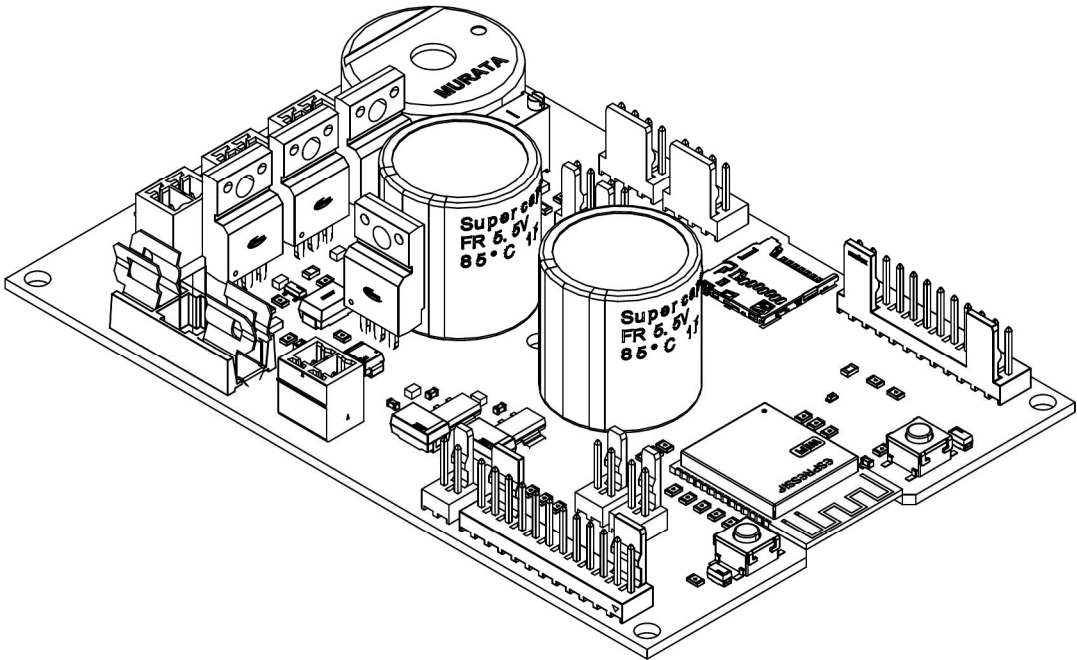
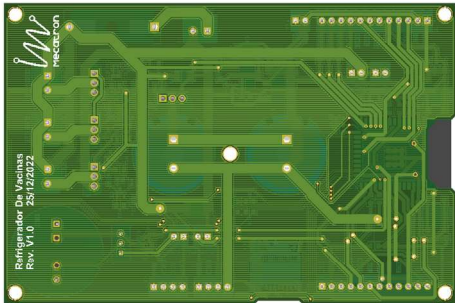
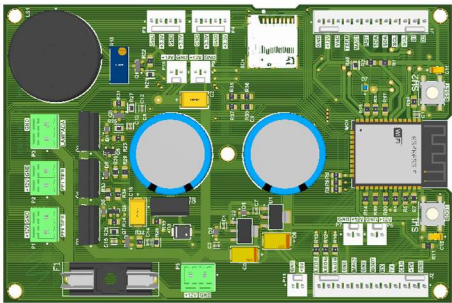
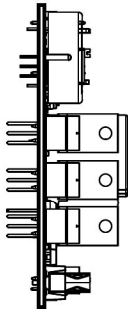


Documentação Geral Reduzida



Esc: 1:2



Esc: 1:1

Title: Refrigerador de Vacinas		Author: Gabriel Canela	
Size: A4		Approved: Nisio José S. Neto	
Unit: mm		Edited: 26/12/2022 10:32	
Date: 03/11/2022		Variant: [No Variations]	
Prj: 0000034		SCH Version: V1.0	
Sheet: 1 of 8		CNPJ: 03.818.375/0001-51	
Company: Mecatron Projetos e Consultoria Junior			



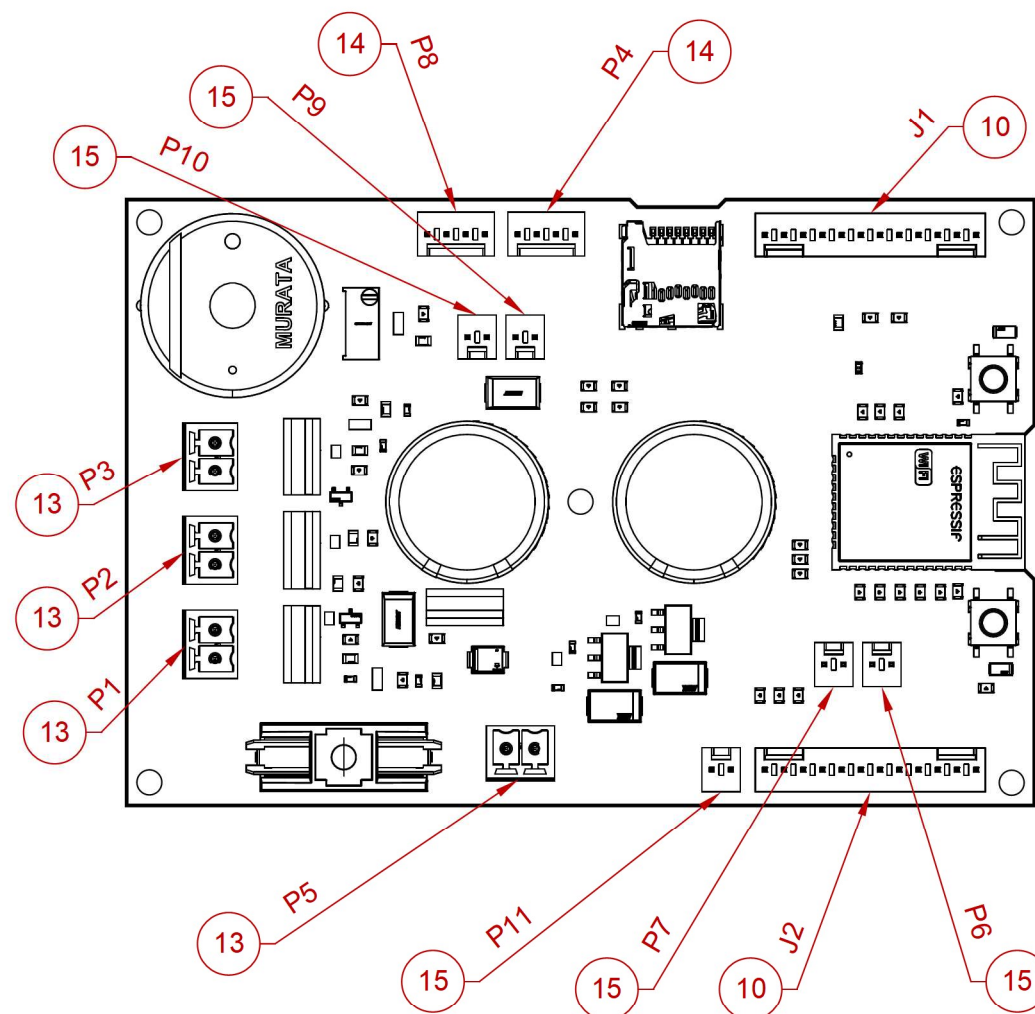
E



Bill Of Materials

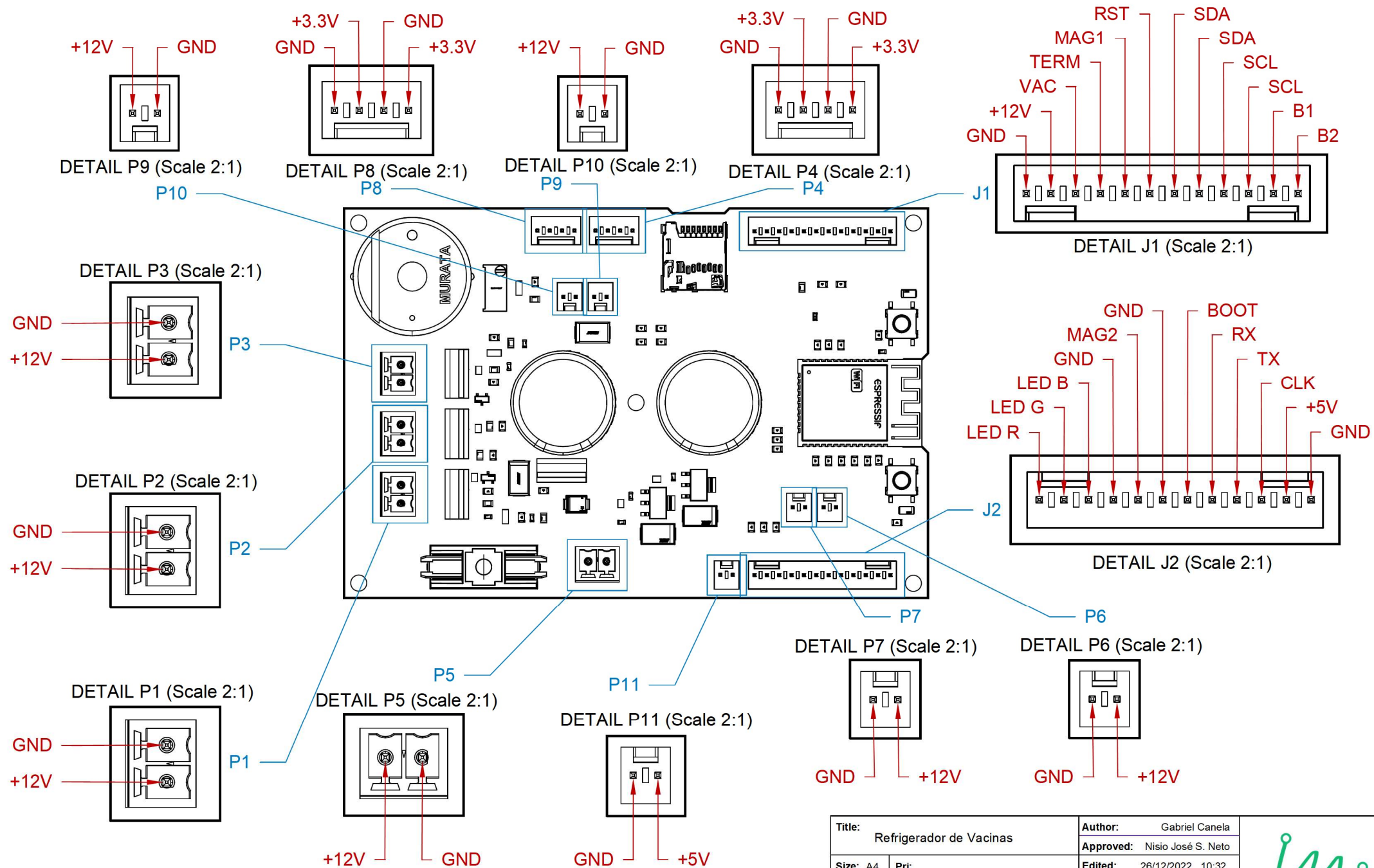
Line #	Designator	Comment	Quantity
1	C1, C3, C4, C7, C12, C13, C14, C15	C0603C471J5GACTU	8
2	C2, C16	T491D476K025AT	2
3	C5, C6	47uF	2
4	C8, C9	Super Capacitor 1F 5.5V, 1F	2
5	C10, C11	TAJA104K035RNJ	2
6	D1, D2, D4, D5, D6	BAT54HT1G	5
7	D3	TVS-15V	1
8	D7	LTST-C191TBKT	1
9	F1	64600001223	1
10	J1, J2	22-27-2121	2
11	LS1	PKB24SPCH3601-B0	1
12	MD1	ESP32-WROOM-32D	1
13	P1, P2, P3, P5	Peltier 1, Peltier 2, Lâmpada, Power	4
14	P4, P8	22-27-2041	2
15	P6, P7, P9, P10, P11	22-27-2021	5
16	Q1, Q7, Q8	BC846BLT1G	3
17	Q2, Q3, Q4, Q9	IRF1404PBF	4
18	Q5, Q6	BC807-40-7-F	2
19	R1, R4, R5, R11, R14, R16, R17, R18, R19, R20, R22, R23, R24, R29, R31, R32, R34, R36, R38	CRCW080510K0FKEA	19
20	R2, R35, R37	CRCW0805100KFKEA	3
21	R3	CRCW08054K70FKEA	1
22	R6, R7, R8, R9	CRCW080547K0FKEA	4
23	R10, R12, R13	CRCW0805220RFKEA	3
24	R15	3296W-1-202LF	1
25	R21, R25, R26, R27, R28, R40	ERA6AEB102V	6
26	R30, R33	ERJ-6GEYJ151V	2
27	R39	CRCW0805220KFKEA	1
28	SD1	503398-1892	1
29	SW1, SW2	BOOT, RESET	2
30	U1	LM1117IMP-3.3/NOPB	1
31	U2	LM1117MPX-5.0/NOPB	1

Indicações de conectores



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Conectores por Sessão



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Company: Mecatron Projetos e Consultoria Junior		CNPJ: 03.818.375/0001-51	



Descrição dos conectores e Portas IO do Processador ESP32

Table

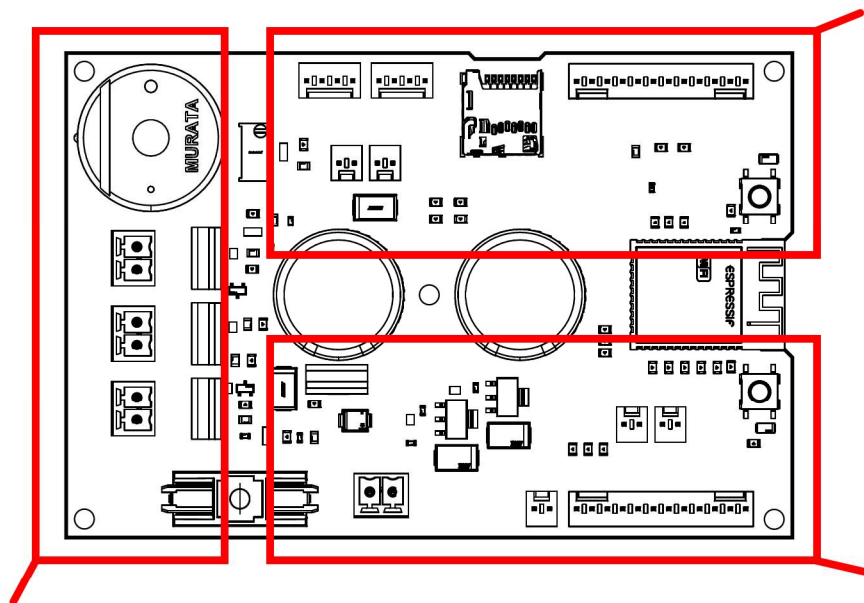
IO Port	Título	Descrição	Indicação PCB
EN	RESET	Reinicia o Centro de Processamento CPU ESP32	RST
IO00	BOOT	Habilita Atualização de Software para CPU ESP32	BOOT
IO02	Cartão MicroSD	Comunicação com MicroSD, porta CS / SS	-
IO04	Controle Peltier	Controle dos módulos peltier, (compatível PWM)	-
IO05	LED	Led RGB, Controle da Cor Azul	LED B
IO12	Sensor Peltier	Sensor de funcionamento Peltier 1	-
IO13	Sensor Peltier	Sensor de funcionamento Peltier 2	-
IO14	Sensor magnético 1	Sensor de porta Aberta	MAG1
IO15	Sensor magnético 2	Sensor de porta Aberta	MAG2
IO16	LED	Led RGB, Controle da Cor Vermelha	LED R
IO17	LED	Led RGB, Controle da Cor Verde	LED G
IO18	Cartão MicroSD	Comunicação com MicroSD, porta CLK	-
IO19	Cartão MicroSD	Comunicação com MicroSD, porta MISO	-
IO21	Comunicação I2C	Conexão SDA	SDA
IO22	Comunicação I2C	Conexão SCL	SCL
IO23	Cartão MicroSD	Comunicação com MicroSD, porta MOSI	-
IO25	Lampada	Controla a Lampada Interna 12V	LAMP
IO26	Sensores termicos	3 Sensores Termicos Conectados em Paralelo	TERM
IO27	Botão externo	Botão Externo 1	B1
IO32	Botão externo	Botão Externo 2	B2
IO33	Buzzer	Controle Buzzer	-
IO34	MicroSwiches	Sensor de Retirada de Vacina	RET
IO35	Sensor Energia	Sensor de Queda de Energia	-

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Indicação de Conexões

Note que o Módulo RTC é externo a PCB e necessita de conexão

Note que os 3 sensores de temperaturas são conectados juntos



Conexões Lado Esquerdo

Objeto	Descrição
Peltier 1	2 fios (+12V, GND)
Peltier 2	2 fios (+12V, GND)
Lâmpada	2 fios (+12V, GND)

Conexões Lado Superior

Objeto	Descrição
Botão do monitor 1	2 fios (+3.3V, B1)
Botão do monitor 2	2 fios (+3.3V, B2)
Display	4 fios (+5V, SCL, SDA, GND)
RTC	4 fios (+3.3V, SCL, SDA, GND)
Sensor magnético porta 1	2 fios (+3.3V, MAG1)
Sensores de Temperatura	3 fios (+3.3V, TEMP, GND)
Sensor de Retirada de Vacina	3 fios (+3.3V, VAC, GND)
Cooler 1	2 fios (+12V, GND)
Cooler 2	2 fios (+12V, GND)

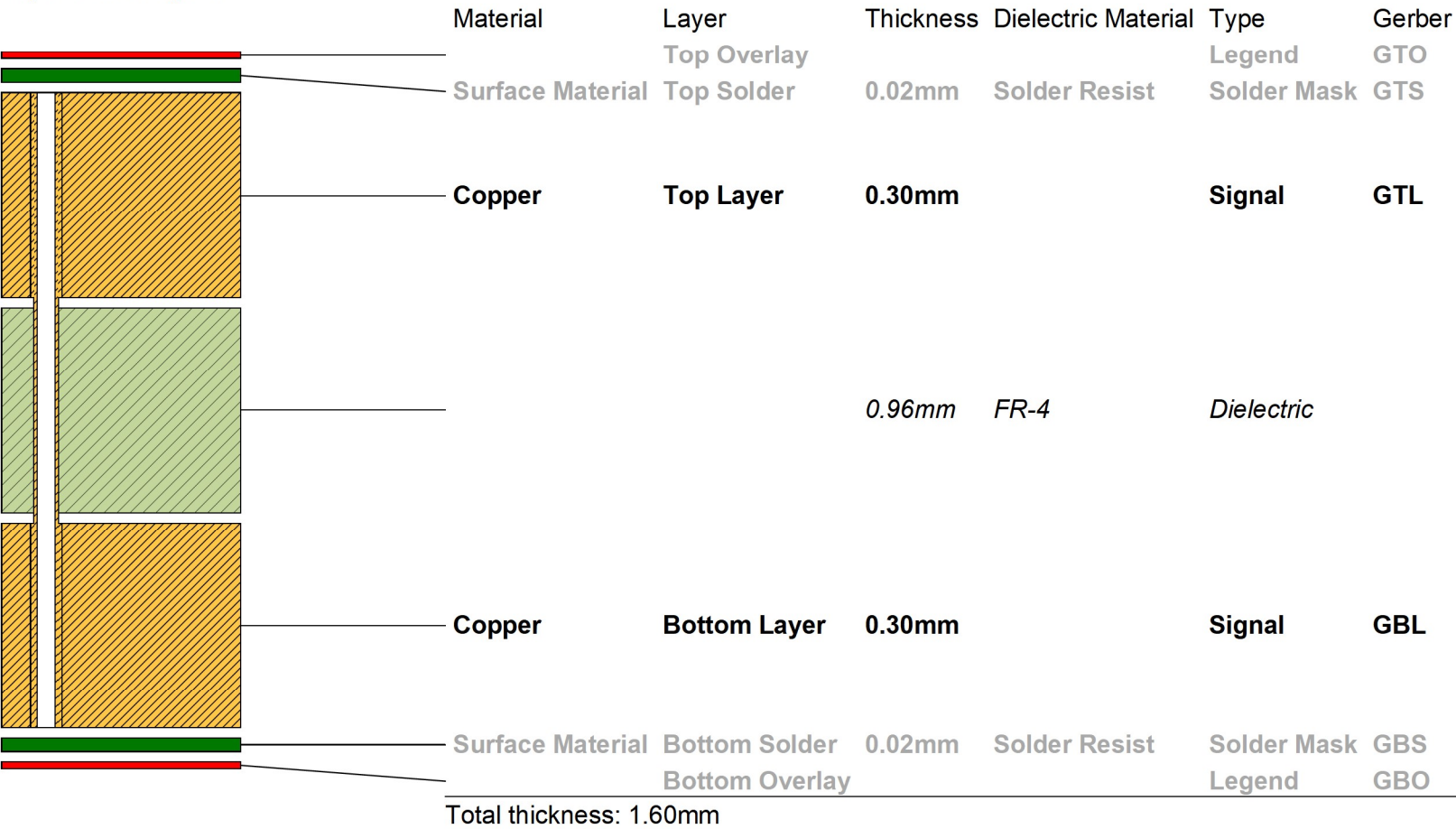
Conexões Lado Inferior

Objeto	Descrição
Sensor Magnético porta 2	2 fios (+3.3V, MAG2)
LED RGB	4 fios (LED R, LED G, LED B, GND)
Entrada de Energia	2 fios (+12V, GND)
Cooler 3	2 fios (+12V, GND)

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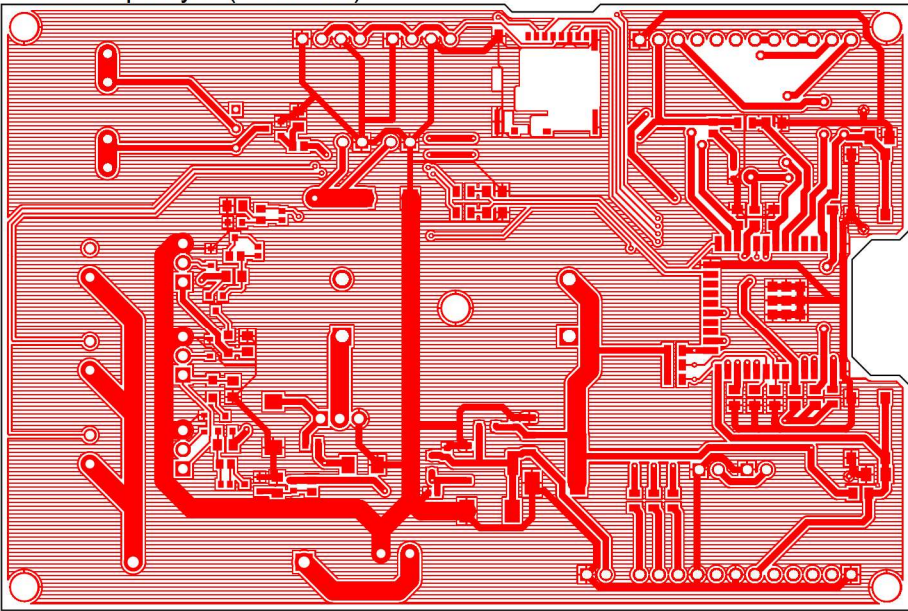
Camadas da Placa de Circuito Impresso

Layer Stack Legend

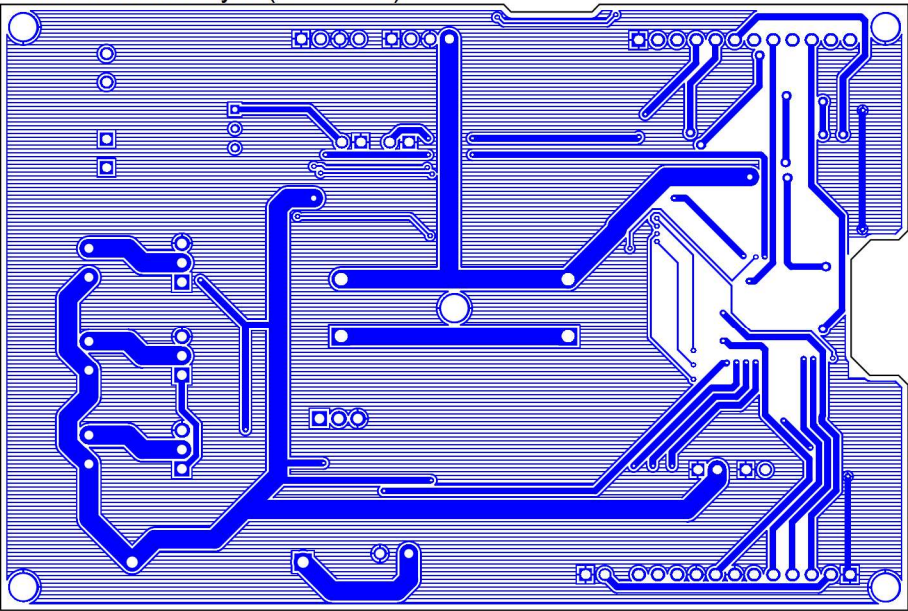


Informações Adicionais De Fabricação

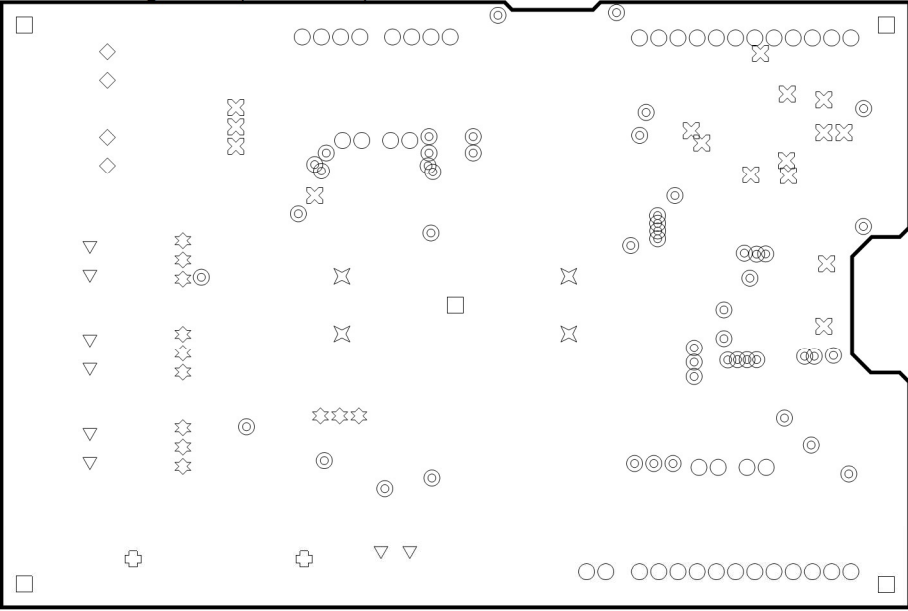
Trilhas Top Layer (Scale 1:1)



Trilhas Bottom Layer (Scale 1:1)



Drill Drawing View (Scale 1:1)



Drill Table				
Symbol	Count	Hole Size	Plated	Hole Tolerance
⊙	50	0.30mm	Есть	
⊗	16	0.70mm	Есть	
◇	4	1.15mm	Есть	
○	42	1.19mm	Есть	
▽	8	1.20mm	Есть	
☆	12	1.30mm	Есть	
⊕	2	1.50mm	Есть	
⊗	4	1.65mm	Есть	
□	5	3.50mm	Нет	
143 Total				

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